DATABASE SYSTEMS ARCHITECTURE DOCUMENT

Banking Database System (BankDB)

Relational Schema Specification & Oracle APEX Implementation Guide

TARGET RDBMS	DOCUMENT VERSION	SCHEMA OWNER	STATUS
Oracle Database / APEX	**1.0.0**	**Student Workspace**	**Production Ready**

1. EXECUTIVE SUMMARY

The Banking Database System (BankDB) is built for Oracle APEX to manage basic everyday banking activities. It securely organizes information for customers, bank accounts, transaction histories, loans, and debit or credit cards. Using clear table relationships and strict database rules, it ensures all bank records stay accurate and organized.

2. ENTITIES & DATA DICTIONARY

Below are the column level definitions, data types, and integrity constraints for the database entities.

2.1 Table: CUSTOMERS

Column Name	Data Type	Key Type	Constraints	Description
`customer_id`	`NUMBER`	PK	`NOT NULL`	Unique system identification number
`first_name`	`VARCHAR2(50)`	—	`NOT NULL`	Customer's primary given name
`last_name`	`VARCHAR2(50)`	—	`NOT NULL`	Customer's surname / family name
`email`	`VARCHAR2(100)`	—	`UNIQUE, NOT NULL`	Contact email address
`phone_number`	`VARCHAR2(20)`	—	`UNIQUE, NOT NULL`	Primary telephone contact number
`national_id`	`VARCHAR2(30)`	—	`UNIQUE, NOT NULL`	Government tax or identity number

2.2 Table: ACCOUNTS

Column Name	Data Type	Key Type	Constraints	Description
`account_id`	`NUMBER`	PK	`NOT NULL`	Internal account record identifier
`customer_id`	`NUMBER`	FK	`NOT NULL`	Refers to `CUSTOMERS(customer_id)`
`account_number`	`VARCHAR2(20)`	—	`UNIQUE, NOT NULL`	Public account number format
`account_type`	`VARCHAR2(20)`	—	`NOT NULL`	Type: CHECKING, SAVINGS, etc.
`balance`	`NUMBER(15,2)`	—	`DEFAULT 0.00`	Current monetary ledger balance

2.3 Table: TRANSACTIONS

Column Name	Data Type	Key Type	Constraints	Description
transaction_id	NUMBER	PK	NOT NULL	Unique ledger record entry ID
account_id	NUMBER	FK	NOT NULL	Refers to ACCOUNTS(account_id)
transaction_type	VARCHAR2(20)	—	NOT NULL	DEPOSIT, WITHDRAWAL, TRANSFER
amount	NUMBER(15,2)	—	NOT NULL	Monetary transaction value
transaction_date	DATE	—	DEFAULT SYSDATE	Execution date & timestamp

2.4 Table: LOANS

Column	Type	Key
loan_id	NUMBER	PK
customer_id	NUMBER	FK
loan_type	VARCHAR2(30)	—
amount	NUMBER(15,2)	—
interest_rate	NUMBER(5,2)	—

2.5 Table: CARDS

Column	Type	Key
card_id	NUMBER	PK
account_id	NUMBER	FK
card_number	VARCHAR2(16)	UNIQUE
card_type	VARCHAR2(10)	—
expiry_date	DATE	—

3. ENTITY RELATIONSHIPS & BUSINESS RULES

- **Customer to Accounts (1 : N):** A customer can own multiple bank accounts (Checking, Savings), but each account is owned by exactly one customer.
- **Account to Transactions (1 : N):** An account maintains a historical log of multiple monetary transactions. Each transaction links back to one parent account.
- **Customer to Loans (1 : N):** A customer can apply for and hold multiple active credit products or loans.
- **Account to Cards (1 : N):** One or more debit or credit cards can be linked directly to an existing bank account.